

Cover Letter

Dear Professor Tawil-Souri and the MCC hiring committee,

I am writing to apply for the Graduate Student Researcher position for Spring 2026. As an MS Computer Engineering student at NYU Tandon with hands-on experience in literature review, data analysis, manuscript preparation, and cross-disciplinary collaboration, I am confident I can contribute to your research from day one.

Why your research. Your work on technology, infrastructure, and politics — particularly how media systems shape lived experience in Palestine, as in *Producing Palestine* (2024) and your writing on digital borders — resonates with my interest in research that uses systematic methods to examine how technology mediates power. I would be eager to support this work.

Industry–academic research collaboration. I co-developed an open-source research project **in collaboration with a Decision Scientist at Merck & Co.**, performing the full research lifecycle: surveying 15+ sources for a reference library, integrating 5 large-scale datasets (47,000+ entities, 2.25M relationships), co-authoring a methods paper through multiple revision cycles, designing a conference presentation, and organizing all data for easy collaborator access.

Independent paper reproduction. I reproduced a published machine learning paper (Nahin et al., arXiv:2510.03950) from scratch — reimplementing 6 core modules, verifying all key claims (exceeding benchmarks in several cases), discovering a methodological error in the original work, and preparing a seminar presentation on my findings.

What I bring. Literature review and citation management; manuscript drafting and copy editing in L^AT_EX; data collection, cleaning, and statistical analysis; publication-quality figures; Git/GitHub and web documentation; and the attention to detail that once caught a parsing error silently invalidating 61,000+ records.

I am available for the full Spring 2026 semester and happy to discuss my qualifications at your convenience.

Best regards,

Shihua Yu

MS Computer Engineering, NYU Tandon

sy4908@nyu.edu

[LinkedIn](#) · [GitHub](#)

Project: github.com/youseihuayu-wonderful/OptimShortestPaths.jl